800 900 1000 1100 1200 1300 1400 1500 1600

Temperature (° C)

40

60

80

100

120

140

160

180

200

220

240

Depth (km)

Young ORCA

NoMelt

Old ORCA

50 100 150 200 250 300 350 400 450

C H 2 O (ppm)

40

60

80

100

120

140

160

180

200

220

240

Bulk

Solid

0 0.02 0.04 0.06 0.08 0.1 0.12 0.14 0.16 0.18 0.2

Melt Fraction (%)

40

60

80

100

120

140

160

180

200

220

240

-0.5 0 0.5 1 1.5 2

log 10 (Grain Size; mm)

40

60

80

100

120

140

160

180

200

220

240

4.1 4.2 4.3 4.4 4.5 4.6 4.7

V S (km/s)

40

60

80

100

120

140

160

180

200

220

240

Predicted

Observed

0 0.005 0.01 0.015 0.02 0.025 0.03 0.035 0.04

Q -1

40

60

80

100

120

140

160

180

200

220

240